International Journal of Information Management 87 (2026) 103003

Contents lists available at ScienceDirect

International Journal of Information Management

journal homepage: www.elsevier.com/locate/ijinfomgt

Artificial intelligence and career development: Concerns and insights from first-generation college students

Xuefei (Nancy) Deng [a,*], Rui Sun [b]

[a] College of Business Administration & Public Policy, California State University, Dominguez Hills, 1000 E. Victoria Street, Carson, CA 90747, United States
[b] College of Business Administration & Public Policy, California State University, Dominguez Hills, 1000 E. Victoria Street, Carson, CA 90747, United States

A R T I C L E I N F O

Keywords:
Artificial Intelligence
GenAI
Career development
AI risk
Career anchor
First Generation College Students
Marginalized workforce

A B S T R A C T

Artificial intelligence (AI) is disrupting workforce and posing an unprecedented threat of job displacement. However, our understanding of AI's role in shaping individual career development is limited. This study provides insights into AI and career development within the context of first-generation college students (FGCSs), a marginalized group that is arguably among the most vulnerable to the career disruption of AI. Employing mixed methods, this exploratory study examines the effects of FGCS status and career anchor on individual concerns about AI's career impact and the perceptions of FGCSs and non-FGCSs regarding their career development. Using survey data from 70 students at a minority-serving public university in the United States, the quantitative analysis shows that FGCS status is positively associated with individual concern about AI's career impact, whereas prior ChatGPT experience is negatively associated with this concern. However, we did not find evidence that a student's career anchor affects their concerns about AI's career impact. Meanwhile, the qualitative analysis revealed four themes that highlight employed FGCSs' reliance on college education to change to a professional career or prepare for entrepreneurship. Our follow-up study revealed four types of individual attitudes toward AI's career impact and suggested that the attitudes are influenced by generational status and career stage. We compare FGCSs and their peers in terms of career stage, career development and attitude toward AI's impact and propose intervention strategies to help FGCSs mitigate AI-related job replacement risks. The study contributes to research on the AI impact on career development of a marginalized population.

1. Introduction

Artificial intelligence (AI) technology is reshaping the workforce, causing anxiety among employees about potential job displacement by AI. Across industry sectors, company CEOs are warning their employees that AI will be used to replace humans in some positions, resulting in employee layoffs. For example, on June 30, 2025, Amazon CEO Andy Jassy predicted that the company's workforce will decline in the next few years as some of the work will be handled by generative AI and AI-powered software agents (Palmer 2025a). Just within four months, on October 28, 2025, Amazon announced its layoff of approximately 14, 000 corporate employees while looking to invest in generative AI (Palmer 2025b). At the technology sector, the accelerated use of AI in automating workloads led Salesforce to cut more than 1000 positions in early 2025 to restructure around AI; by June 2025, AI is doing up to 50 % of the work at the company, according to CEO Marc Benioff (Subin, 2025). At the retail sector, Target recently announced eliminating about 1800 corporate jobs (approximately 8 % of its corporate workforce), the largest round of layoffs at the company in a decade (Repko, 2025). Walmart CEO Doug McMillon said that "It's very clear that AI is going to change literally every job" (Nassauer, 2025).

AI and its latest development of GenAI using LLM have posed an unprecedented threat to job displacement in workplaces. According to a Brookings Report, unlike previous automation technologies that primarily affected routine, blue collar work, GenAI is likely to disrupt a different array of "cognitive" and "nonroutine" tasks, especially in middle- to higher-paid professions (Kinder et al., 2024). For example, a survey of more than 600 consultants at Boston Consulting Group shows that GenAI performs better on a consulting task than a junior (novice) consultant (Bowen & Watson, 2024). Scholars across multiple disciplines have highlighted the risks of AI-driven deskilling (Bowen & Watson, 2024; Dwivedi et al., 2023).

AI is capable of automating human tasks and augmenting human work, reshaping business operations, employees' roles and customers'

* Corresponding author.
E-mail addresses: ndeng@csudh.edu (X.(N. Deng), rsun@csudh.edu (R. Sun).

https://doi.org/10.1016/j.ijinfomgt.2025.103003
Received 29 November 2024; Received in revised form 9 November 2025; Accepted 14 November 2025
Available online 21 November 2025

places in service interactions (Belk et al., 2023). For example, in the financial industry, AI has been used to predict loan default and produced more accurate results than crowd investors, improving the welfare of both investors and borrowers (Fu et al., 2021). In the hospitality industry, service robots are found to possess more sophisticated types of AI such as feeling AI and thinking AI, which could increase spending and loyalty intention through customers' positive emotions (Schepers et al., 2022) and help firms to achieve cost-effective service excellence (Wirtz et al., 2023). The recent advancements of AI based on large language models (LLM) have unleased new, human-like capabilities. As a subset of AI technologies, generative AI (GenAI) can generate seemingly new, meaningful content such as text, images, or audio based on the training data (Feuerriegel et al., 2024). The content generated by GenAI, in many times indistinguishable from human output, can be applied across a wide range of knowledge-intensive contexts.

As AI increasingly reshapes the workforce, it is causing career-related concerns not only among employees but also among college students, the future workforce. With rising concerns about AI displacing human workers, it becomes critical for higher education institutions and organizations to understand individuals' concerns about AI's impact on their careers. Equally important is examining the effect of career anchors on the AI concerns, as the career anchor is a key individual characteristic associated with career choice and development (Schein, 1985; Schein, 1996).

A marginalized group that is arguably among the most vulnerable to the career disruption of AI is first-generation college students (FGCSs). FGCSs refer to students who are enrolled in post-secondary education and whose parents do not have such a college educational experience (Nunez & Cuccaro-Alamin, 1998; Redford & Hoyer, 2017). Compared to their continuing-generation college student (CGCS) peers (e.g., students whose parents have college degrees), FGCSs are twice as likely to leave college without a degree (Chen & Carroll, 2005) and their FGCS status negatively affects not only the graduates' career decision-making but also their familiarity with corporate expectations and preparedness for the corporate sector (Hirudayaraj & McLean, 2018). Nevertheless, the career transition and success of FGCSs remain important phenomena. In the United States, FGCSs account for approximately 25 % of the undergraduate population (Dika & D'Amico, 2016). The advancements of AI and GenAI have given rise to new threats to job displacement across various sectors. If we are to build a nation where everyone pursuing a higher education can experience equal career opportunities and achieve success, higher education institutions and organizations must not ignore FGCSs' concerns and career challenges in the age of AI.

To address these phenomena and explore the profound influence of GenAI on individuals and society, this exploratory study investigates individual concerns about AI's impact on career development from the perspectives of the FGCSs in a marginalized context. Specifically, the research questions (RQs) are:

- RQ1: Does a student's FGCS status affect their concerns about AI's impact on their career?
- RQ2: Does a student's career anchor affect their concerns about AI's impact on their career?
- RQ3: How do FGCSs and their non-FGCS counterparts perceive their career development?

To answer these questions, the study draws upon the literature on FGCS graduates, career anchor theory, and AI and career development. We aim to capture the nuanced factors that inform and shape FGCSs' career development, thereby enriching our understanding of their perceptions of AI's impact on their career and professional growth.

This study adopts a mixed-methods approach that includes both quantitative and qualitative data collection and analysis techniques, allowing for a more comprehensive understanding of the research topic by leveraging the strengths of each method (Mingers, 2001; Mingers et al., 2013). Quantitative methods are used to answer the first two research questions (RQs 1–2); and a qualitative method is employed to address the third research question (RQ3). Moreover, to enhance the qualitative analysis by learning more about FGCSs' attitudes toward AI's impact on their career development, we conducted a follow-up study, offering additional insights into the complex phenomena.

The remaining sections of this article are organized as follows. Section 2 reviews the literature on FGCS graduates in the workplace, career anchor, and AI and career development, providing the theoretical background for the study. Section 3 describes the research methods used for collecting and analyzing the data. The quantitative analysis and results are reported in Section 4, followed by the qualitative analysis and findings in Section 5. Section 6 presents a follow-up study to enhance the qualitative analysis; Section 7 discusses the theoretical contributions and practical implications of the study; and Section 8 concludes the study with a summary of findings, limitations of the study, and suggestions for future research.

2. Literature review

2.1. FGCS graduates in the workplace

Research has identified inhibitors particularly associated with FGCSs' persistence and success in higher education, including low household income, full-time work demands, and lack of social support (Engle & Tinto, 2008; Lohfink & Paulsen, 2005; Nunez & Cuccaro-Alamin, 1998). These inhibitors remain as FGCSs transition from college to workplace. According to Chen and Carroll (2005), FGCSs struggle to choose a major—partly because their parents are unable to offer knowledgeable guidance about the process and the prospect of a career. This lack of support and career guidance is evidenced in an empirical study (Toyokawa & DeWald, 2020): FGCSs scored higher on lack of support and lack of time and financial resources than CGCSs in deciding on their career. The lack of a college education in the families of the FGCSs negatively affected not only the graduates' career decision-making but also their preparedness for the corporate world (Hirudayaraj & McLean, 2018). As a result, FGCS graduates transitioning from college to workplace are likely to be at a disadvantage compared to their CGCS peers.

However, FGCS status has received limited attention in the literature on postgraduate careers and the workforce until recently. Researchers have started to explore the factors that influence FGCSs' career considerations and career success. For example, Olson (2014) suggested using social cognitive career theory (SCCT) as a framework to support individuals during the early stages of their careers by exploring the individual's self-efficacy, outcome expectations, and personal goals. This approach provides an effective tool for career counselors who work with FGCS graduates as they embrace opportunities, face obstacles, and explore options. Similarly, guided by SCCT, Tate et al. (2015) revealed three major domains – external influences on the career development process, understanding of the career development process, and self-concept. For FGCSs, one external influence on their career development is the social capital that some of them have accumulated in their lived experiences. As revealed in Deng et al. (2022), both the social capital and the technical capital of FGCSs emerged as positive and direct influences on their decision to choose a career in information technology (IT).

While the literature on FCGSs' career choices and development remains limited, the literature offers a glimpse into the challenges that FGCSs encounter in entering competitive and rapidly changing career fields. As the future of work and career is becoming unpredictable with the rapid penetration of AI in workplaces, FGCSs are undoubtedly a population that is among the most vulnerable to the emerging disruptions in career fields that are being shaped by the new technology. Thus, this study seeks to investigate the concerns and career considerations of this vulnerable workforce in the age of AI. To achieve this objective, we draw upon the career anchor theory originally developed by Schein

(1978), as it provides a robust framework for analyzing career motivations and decisions at the individual level.

2.2. Career anchor theory

Career anchor theory was proposed by Edgar Schein (1978). According to this theory, an individual's career is "anchored" in a set of needs and motivations that the individual tries to satisfy through his or her work and its anticipated benefits. Schein (1985) identified eight career anchors: technical/functional competence, managerial competence, autonomy/independence, security/stability or organizational identity, entrepreneurial creativity, service/dedication to a cause, pure challenge and lifestyle integration.

First, the ***technical/functional competence*** career anchor refers to the extent to which workers value the perception of their technical and functional competence as a motivation for professional activities. In contrast, the second career anchor, ***managerial competence***, refers to the motivation of a worker to take on managerial responsibilities—or "climb the ladder" of organizational leadership positions. Third, the ***autonomy/independence*** career anchor refers to the extent to which a worker values self-reliance and individual decision-making about work activities. Fourth, the ***security/stability*** career anchor initially referred to employment security—or the likelihood that the worker would be able to continue in their current employment (Schein, 1978)—but evolved to focus more on "employability" security, i.e., developing and maintaining marketable skills (Schein, 1996).

Fifth, the ***entrepreneurial creativity*** anchor refers to the extent to which a worker is drawn to the idea that they can develop their own business or contribute to new products or services that will create value in the economy (Schein, 1996). This career anchor can be divided into two aspects: entrepreneurship and creativity. Sixth, the ***service/-dedication*** career anchor refers to the extent to which a worker is motivated by a desire to do something meaningful for society, as well as to earn an income (Schein, 1996). Seventh, the ***pure challenge*** career anchor refers to the extent to which a worker is motivated by addressing challenges and problems by using their intellect and creativity (Schein, 1996). Lastly, the ***lifestyle integration*** career anchor refers to the extent to which a worker is motivated by integrating their personal and professional activities into a balanced life (Schein, 1996).

According to Schein (2006), individuals have only one dominant profile that guides and constrains their entire career. However, this notion contrasts with Feldman and Bolino (1996) as they analyzed Schein's, (1978) study and noted that about one third of respondents had a "multiple" career anchor profile, suggesting the presence of "primary" and "secondary" anchors. Furthermore, to explain why some individuals have several dominant career anchors, Wils et al. (2014) grouped the anchors proposed by Schein (1985) into four large clusters of values characterized by opposed pairs: bureaucratic self-concept (BSC) opposed to the protean self-concept (PSC), and the careerist self-concept (CSC) opposed to the social self-concept (SSC). BSC is linked to the security/stability and technical/functional competence anchors, while PSC is associated with the pure challenge anchor. In addition, CSC is associated with managerial competence, identity, independence and entrepreneurship anchors, whereas SSC is associated with the service/dedication anchor. This new career value structure model was validated using a sample of 240 employees and 155 managers in a health care organization (Wils et al., 2014).

In the Information Systems (IS) research studying IT careers, scholars have adopted career anchor theory (Taylor & Joshi, 2019) and change in career anchors over time in the same person as they become more experienced (Chang et al., 2011). For example, IT workers who remain in technology-related positions at traditional firms become more interested in managerial positions and in geographic security and value higher levels of autonomy (Chang et al., 2011). On technology-enabled digital labor platforms, certain career anchors, such as managerial competence, and security/stability no longer exist (Taylor & Joshi, 2019). As AI technologies are transforming our workplaces and posing threats to job displacement, some career anchors established in the traditional work settings may no longer exist. In this regard, we review recent studies on AI and career development for further insights.

2.3. AI and career development

Recent advances in the field of AI (e.g., GenAI) not only offer new opportunities for individuals' career development but also cause several concerns. As Dwivedi et al. (2021) argued, AI-related challenges could encompass social, economic, ethical, political, legal, managerial, technological, and data dimensions. On one hand, GenAI may have extensive implications in business sectors such as banking (Dwivedi et al., 2023), travel and tourism (Wong et al., 2023), and health care (The Lancet Regional Health – Europe, 2023), as well as in business activities such as marketing and management (Kshetri et al., 2023). On the other hand, the increasing reliance on GenAI has raised questions about human-AI relations and the disruption of the labor market. Therefore, there is a need to understand GenAI applications and their positive and negative consequences for consumers, companies, and society.

Inspired by Belanche et al. (2020) of service robots' implementation in service industries, Belanche et al. (2024) proposed a three-part conceptual model comprised of AI design, customers, and the service encounter and outlined critical aspects, concerns, and threats of AI implementation related to each of these three parts. One of the concerns regarding the service encounter part is employment replacement, particularly for low-skilled workers, and social inequalities between developed countries or regions with more AI-related high-skill jobs and those with economies based on low-skill jobs (Belanche et al., 2024). Another concern is that AI lacks emotion, consciousness, and life purpose. This concern is particularly relevant in service industries where an automated social presence (ASP) – "the extent to which machines (e.g., robots) make consumers feel that they are in the company of another social entity" (Van Doorn et al. 2017, p. 44) – affects consumers' perceptions of being served, understood, empowered, and connected regarding their AI experiences (Flavián et al., 2024). More interestingly, the ASP's effect on consumers' feelings of being connected is stronger for those with higher needs for social interaction (Flavián et al., 2024). Therefore, because AI has been integrated into various business sectors, it is important to examine the new job roles or career paths emerging for both low- and high-skill workers and for sectors requiring high levels of emotional intelligence (Belanche et al., 2024).

Scholars have shared the view that AI has a double-edged sword effect on career development (Pandya & Wang, 2024): while AI technology helps streamline career development processes in organizations, it can also disrupt a workplace and cause job insecurity. In the new landscape of professional careers shaped by the latest AI technologies, the career anchors of FGCSs and their perceptions of AI's impact on their career remain important questions. As prior research has suggested, understanding the self-efficacy, outcome expectations, and personal goals of FGCS graduates is essential to explaining their early-stage career development (Olson, 2014; Tate et al., 2015). In particular, the outcome expectations for job opportunities in a professional field have a significant, positive effect on FGCSs' career choices, such as choosing an IT career (Deng et al., 2022). Similarly, in the context of AI-transformed workplace, understanding the nuances of the FGCSs' career anchors and their perceptions of AI's impact on their career could provide insights and offer useful guidance for FGCSs to better prepare them for their future career.

3. Methods

This study employs mixed research methods that include both quantitative and qualitative data collection and analysis techniques. The first two research questions (RQs 1–2) are addressed through quantitative analysis using *t*-tests, factor analysis, and multivariate regressions on numerical survey responses and the last research question (RQ3) is addressed through thematic analysis of open-ended survey questions, a qualitative method for identifying, analyzing, and reporting certain themes or patterns on narrative data (Braun & Clarke, 2006; Clarke et al., 2015). While quantitative analysis identifies broader patterns with statistical significance, qualitative analysis provides more depth and rich contextualization. Therefore, mixed methods allow for a more comprehensive understanding of the research topic by leveraging the strengths of each method (Mingers, 2001; Mingers et al., 2013).

3.1. Research site

The study uses a sample of undergraduate and graduate students in a business college at a public university in the United States. The university is known as a minority-serving institution, with 68 % of students being Hispanic or Latino, 12 % Black or African American, and 8 % Asian or Pacific Islander, based on 2025 university data. In the United States, minority-serving institutions are institutions of higher education that enroll a high percentage of minority students such as African American, American Indian, Hispanic/Latino, and Pacific Islander (US Department of Education, 2021). Most students in this university come from underserved local communities. Underserved students are students who do not receive educational and career planning opportunities and resources similar to those of other students in the academic pipeline (ACT Inc, 2014). Underserved students possess at least one of the following characteristics: (1) racial or ethnic minority status; (2) low household income; or (3) being a first-generation college student (i.e., highest parental education level is high school or less) (ACT Inc, 2014). In 2022, approximately 92 % of all undergraduate students enrolled in this university were considered underserved (they met at least one of the ethnicity, income, and generation criteria), among which, 30 % possessed all three characteristics (i.e., minority, low-income, and first-generation).

3.2. Data and sample

The data were collected through a student survey conducted in May-July 2023. The researchers invited instructors in the business college of this university to participate in the study by adapting and implementing a ChatGPT learning activity in their courses and disseminating a survey to their students after the students had completed the ChatGPT activity. This business college offers both business administration and public administration degrees. Four instructors voluntarily agreed to participate and distributed the survey to their students in the six courses they were teaching.

The survey was created by the researchers and hosted on a university-licensed online survey platform. It consisted of 28 multiple-choice or open-ended questions. Key survey questions included the following: “What do you consider a successful career? Please indicate your agreement to the following statements.” “What career (job and industry) will you be interested in pursuing after your graduation and why? Please explain.” “How likely do you think the following occupations will be replaced by AI (e.g., ChatGPT)?” In addition, the survey included questions on the students’ demographic and socioeconomic characteristics such as age, gender, race or ethnicity, employment status, and FGCS status. The survey was approved by the Internal Review Board (IRB) of the university to ensure that the research was conducted ethically.

A total of 70 students (an overall response rate of 40 %) responded to the survey anonymously. Among the 70 participants, 53 indicated that they had no prior experience with ChatGPT. The 17 students who had used ChatGPT mentioned that they had used it in another class (n = 8), for work (n = 1), during job applications (n = 4), or for other personal interests (n = 4). Most student participants (60 %) were FGCSs while majority of the student participants were employed, including full-time employed (58.6 %) or part-time employed (22.9 %). Student participants represented diverse ethnic backgrounds: 51.4 % Hispanic, 20 % Black, and 10 % Asian or Pacific Islander. Female students comprised 55.7 % of the sample, and most students were between 18 and 39 years’ old. Student participants were majoring in Business Administration (68.6 %) and Public Administration (28.6 %). In general, the sample statistics are consistent with the university student demographics in terms of ethnicity, employment status, and gender, while having a higher representation of FGCSs. This is a convenience sample. Table 1 presents the descriptive statistics of sample demographics.

4. Quantitative analysis

4.1. The influence of FGCS status on students’ concerns about AI’s impact on their career

To answer RQ1, whether a student’s FGCS status affects their concerns about AI’s impact on their career, we first compared the perceptions of FGCSs and non-FGCSs regarding which occupations will likely be replaced by AI. Because our study participants are students from a business and public policy college, we listed 19 common occupations for business and public administration graduates and asked the survey respondents to answer the question “How likely do you think the following occupations will be replaced by generative AI (e.g., ChatGPT)?” on a 5-point Likert scale (1 = Not Likely, 3 = Neutral, and 5 = Extremely Likely). When the median response to an occupation was greater than 3, we considered it an occupation at high risk of being displaced by GenAI. Similarly, median responses of 3 were coded as medium risk, and median responses smaller than 3 were coded as low risk. Table 2 presents the perceived risks of the 19 occupations by FGCSs and non-FGCSs.

In the presence of GenAI, job replacement concerns differ by professional fields. Participants in our study shared a common concern that positions such as accountants, analysts, customer service, and computer programmers are subject to high risks of being replaced by GenAI. Meanwhile, positions that require direct interactions with people, such as correction officers, police officers, social workers, lawyers, human

Table 1
Descriptive statistics of sample demographics (n = 70).

	Frequency	Percent	University Data, 2022–2023
By FGCS Status			
FGCS	42	60.0 %	47.3 %
Non-FGCS	22	31.4 %	
Prefer not to answer	6	8.6 %	
By Employment Status			
Employed full-time	41	58.6 %	74 %
Employed part-time	16	22.9 %	
Not employed	13	18.6 %	
By Ethnicity			
Asian or Pacific Islander	7	10.0 %	7.5 %
Black or African American	14	20.0 %	10.9 %
Hispanic or Latino	36	51.4 %	69.4 %
White / Caucasian	5	7.1 %	5.8 %
Other	2	2.9 %	
Prefer not to answer	6	8.6 %	
By Gender			
Women	39	55.7 %	62.4 %
Men	27	38.6 %	37.5 %
Prefer not to answer	4	5.7 %	
By Major			
Business Administration	48	68.6 %	
Public Administration	20	28.6 %	
Other	2	2.9 %	
By Age			
18–21	13	18.6 %	N/A
22–29	25	35.7 %	N/A
30–39	22	31.4 %	N/A
40–49	7	10 %	N/A
50–59	2	2.9 %	N/A
Prefer not to answer	1	1.4 %	

Table 2
Perceived risk of GenAI replacing occupations in business and public administration: by FGCS status.

	All (n = 70)		FGCS (n = 42)	non-FGCS (n = 22)	*t*-tests: FGCS vs. non-FGCS
	Median	Mean	Mean	Mean	Direction of Effect and Significance
High-risk positions (Median > 3)					
(1)-Accountants	4	3.5	3.5	3.4	
(2)-Business Analysts	4	3.8	4	3.5	(+)*
(3)-Computer Programmers	4	3.6	3.9	3.1	(+)***
(4)-Customer Service	4	3.2	3.3	3.1	
(5)-Data Analysts	4	3.7	4.1	3.1	(+)***
(6)-Market Research Analysts	4	3.7	4.2	3	(+)***
(7)-Policy Analysts	3.5	3.2	3.3	3	
Medium-risk positions (Median = 3)					
(1)-Cybersecurity Specialists	3	3.1	3.3	2.7	(+)*
(2)-Financial Advisors	3	3.1	3.4	2.7	(+)**
(3)-Logistics Coordinator or Specialists	3	3.2	3.3	3	
(4)-Office Administration	3	3	3	3	
(5)-Sales (e.g., real estate agent, car sales)	3	2.6	2.5	3	(-)*
Low-risk positions (Median < 3)					
(1)-Correctional Officers	2	2.1	2.1	2.2	
(2)-Human Resource Specialists	2	2.6	2.7	2.6	
(3)-Lawyers/paralegals	2	2.1	2.1	2.2	
(4)-Police Officers	1	1.8	1.7	2.1	
(5)-Social Workers	1	2.1	1.9	2.6	(-)**
(6)-Teachers	2	2.5	2.5	2.5	

Note. Six respondents did not indicate their FGCS status. * $p < 0.1$, ** $p < 0.05$. *** $p < 0.01$.

resource specialists, and teachers, are in the low-risk category of job displacement by GenAI. From their perspective, positions in cybersecurity, financial advising, logistics coordination, office administration and sales/marketing are subject to job displacement risk at the medium level.

When we compared the perceived GenAI impact of FGCSs and non-FGCSs, the *t*-test results show that on average, FGCSs perceived higher levels of risk, compared to non-FGCSs, in relation to four of the seven high-risk positions and two of the five medium-risk positions. The four high-risk positions are business analysts, computer programmers, data analysts, and market research analysts. The two medium-risk positions are cybersecurity specialists and financial advisors. On the other hand, FGCSs perceived lower levels of risk, compared to non-FGCSs, in relation to one medium-risk position (sales) and one low-risk position (social workers).

Next, we performed multivariate regressions to test the relationship between FGCS status and individual concerns about AI's career impact. The dependent variable, the concern about AI's impact on a career, is measured by the survey question "I am worried about the impact of AI tools (e.g., ChatGPT) on my career or potential career" (1 = Strongly Disagree; 5 = Strongly Agree). The independent variable is FGCS (1 = Yes, 0 = No) and the control variable is prior ChatGPT experience (1 = Yes, 0 = No). Considering that the dependent variable is an ordinal variable with ranked categories and the independent and control variables are dichotomous, ordered logistic regression is applicable. However, this regression technique generally requires larger sample sizes to achieve reliable results. Thus, we use an ordinary least squares (OLS) regression as a baseline model for comparisons. Table 3 presents the results of the OLS regression (Model 1) and the ordered logistic regression (Model 2).

As shown in Table 3, both models indicate that FGCS status is positively associated with the concern about AI's impact on careers, controlling for students' prior ChatGPT experience (statistically significant at the 0.05 level). Compared to non-FGCS students, FGCS students are more worried about the impact of AI on their career or potential career. In addition, the OLS model suggests that students with prior ChatGPT experiences are less worried about AI's impact on their career, compared to students who had not used ChatGPT before (statistically significant at the 0.1 level). However, the effect of prior ChatGPT experience is not statistically significant in the ordered logistic regression model.

Table 3
Regression results regarding RQ1.

DV= Concern about AI's impact on career	**Model (1) OLS**	**Model (2) Ordered Logistic**
FGCS	0.702**	0.966**
	(0.333)	(0.484)
Prior ChatGPT	-0.629*	-0.841
	(0.365)	(0.526)
Constant	2.978***	
	(0.276)	
Observations	64	64
R-Square	0.10	0.10

Note. Standard errors are in parentheses. Coefficients (log odds) are reported in the ordered logistic regression. * $p < 0.1$, ** $p < 0.05$. *** $p < 0.01$.

4.2. *The influence of career anchor on students' concerns about AI's impact on their career*

To answer RQ2, whether a student's career anchor affects their concern about AI's impact on their career, we performed multivariate regressions using the concern about AI's impact on career as the dependent variable and career anchor as the independent variable, controlling for FGCS status and students' prior ChatGPT experience.

The independent variable, career anchor, is measured by the survey question "What do you consider a successful career? Please indicate your agreement to the following statements: (1) I can develop my technical or functional skills to a very high level of competence; (2) I become a general manager in some organizations; (3) I achieve complete autonomy and freedom; (4) My job gives me a sense of security and stability; (5) I have succeeded in starting up and building my own business; (6) I have succeeded in my creating or building something that is entirely my own product or idea; (7) I have a feeling of having made a real contribution to the welfare of society; (8) I face and overcome very difficult challenges; and (9) I have been able to balance my personal, family, and career requirements." The responses are measured on a 5-point Likert scale (1 = Strongly Disagree; 5 = Strongly Agree). The nine statements or items correspond to the eight career anchors proposed by Schein (1985), except that the entrepreneurial creativity anchor is represented by two items (Items 5 and 6).

A factor analysis (principal component analysis) on career anchor yielded two components: the first includes seven of the nine items (Items 1, 2, 3, 4, 7, 8, and 9) and explains 46 % of the variance; the second component includes the remaining two items (Items 5 and 6) and explains 16 % of the variance. Thus, we loaded the seven items onto the same factor (eigenvalue of 4.16) and created a 7-item additive index, *Career Anchor-1*. The Cronbach's alpha for these seven items is 0.858, indicating strong internal consistency. The variable, *Career Anchor-1* (Mean= 4.01; Standard Deviation = 0.70) ranges from 1.14 to 5, a higher index score indicating a higher level of agreement with the relevant statements regarding a successful career. We loaded the other two items (Items 5 and 6) onto the same factor to create a 2-item additive index (eigenvalue of 1.44), *Career Anchor-2*. The Cronbach's alpha for these two items is 0.814, indicating strong internal consistency. The variable *Career Anchor-2* (Mean= 3.48; Standard Deviation = 1.03) ranges from 1 to 5, a higher index score indicating a higher level of agreement with the relevant statements regarding a successful career. Table 4 shows the descriptive statistics for all variables included in the regression models.

Table 5 reports the regression results regarding RQ2. The results of both the OLS regression (Model 1) and the ordered logistic regression (Model 2) indicate that FGCS status is positively associated with individual concern about AI's career impact and prior ChatGPT experience is negatively associated with it, which is consistent with the results of RQ1. However, neither category of the career anchor demonstrated a statistically significant relationship with individual concern about AI's career impact.

The results of the quantitative analysis suggest that FGCSs are more concerned about AI's impact on their career than non-FGCSs, while students with prior ChatGPT experience are less concerned about it. However, we did not find evidence that students' career anchors affect their concerns about AI's impact on their career. To learn more a student's career anchor, we performed *t*-tests to examine whether students' career anchors vary by FGCS status. Table 6 reports the average score (mean) and standard deviation for each of the nine items measuring career anchors within the full sample (n = 70), sub-sample of FGCSs (n = 42), and sub-sample of non-FGCSs (n = 22). It also shows the results of *t*-tests comparing the average scores of FGCSs with those of non-FGCSs.

Using the 0.05 statistical significance level, the *t*-test results suggest that on average, FGCSs have a higher level of agreement to the statements that a successful career means entrepreneurship and creativity: "I have succeeded in starting up and building my own business" (Item 5) and "I have succeeded in my creating or building something that is entirely my own product or idea" (Item 6). Using the 0.1 statistical significance level, the *t*-test results also suggest that on average, FGCSs have a higher level of agreement to the statement that a successful career means autonomy or independence: "I achieve complete autonomy and freedom" (Item 3). In summary, statistical evidence indicates that only three of the nine career anchors vary by the FGCS status of the students. To further understand these results, we conducted a qualitative analysis to explore how FGCSs and their non-FGCS counterparts perceive their career development.

Table 4
Descriptive statistics of variables for RQ2.

	N	Minimum	Maximum	Mean	Std. Deviation
Concern about AI's impact on career	69	1	5	3.25	1.29
FGCS	64	0	1	0.66	0.48
Prior ChatGPT	70	0	1	0.24	0.43
Career Anchor 1	70	1.14	5	4.01	0.70
Career Anchor 2	70	1	5	3.48	1.03

Table 5
Regression results regarding RQ2.

DV= Concern about AI's impact on career	Model (1) OLS	Model (2) Ordered Logistic
FGCS	0.590*	0.818*
	(0.338)	(0.491)
Prior ChatGPT	-0.833**	-1.142**
	(0.386)	(0.567)
Career Anchor 1	0.073	0.208
	(0.250)	(0.361)
Career Anchor 2	0.247	0.332
	(0.170)	(0.246)
Constant	1.935*	
	(1.007)	
Observations	64	64
R-Square	0.138	0.138

Note. Standard errors are in parentheses. Coefficients (log odds) are reported in the ordered logistic regression. * $p < 0.1$, ** $p < 0.05$.

5. Qualitative analysis

5.1. Qualitative data coding and analytical procedure

We used thematic analysis to interpret the narrative data and answer RQ3 about how FGCSs and their non-FGCS counterparts perceive their career development. Thematic analysis is a qualitative method for identifying, analyzing, and reporting certain themes or patterns within qualitative data (Braun & Clarke, 2006; Clarke et al., 2015). It involves systematically coding data, recognizing patterns, and organizing these patterns into meaningful themes that address the research question. Following the guidelines articulated by Braun & Clarke, (2006), we conducted thematic analysis to interpret student responses to two open-ended survey questions: "If you are employed, what do you do at work during a typical day?" "What career (job and industry) will you be interested in pursuing after your graduation and why? Please explain." To examine their personal experiences and the potential influence of their background, we also utilized the demographic data collected from the survey.

Our data analysis followed the four-step guideline proposed by Braun & Clarke, (2006). First, informed by prior research on FGCSs and career anchors, we performed an open coding of student narratives to identify an initial pool of codes, referred to as first-order codes. Initial codes included switching careers, career advancement, career transition, and career interest as the driver of a new career. Second, we sorted and grouped the codes into higher-order categories based on the shared commonalities (such as their employment status) and collated each code's data extracts to its corresponding theme. Next, we iteratively refined the codes by screening all collated extracts for each theme and made the necessary revisions. Finally, we checked for missing codes and organized the subgroup codes in a hierarchical structure.

Table 7 illustrates our qualitative data coding and analysis process. Using two responses as examples, we explain the process from revealing the first-order codes to grouping them into higher-order categories and final themes. As shown in the first response, two concepts—switching careers and career interest—were revealed to be associated with the career development plan of the respondent, who is a full-time Transportation Security Administration (TSA) employee. In the second response, the respondent expressed his/her career goal of becoming a social media manager while currently working part-time at a burger restaurant. The two same concepts of switching careers and career interest were revealed. Thus, the first-order codes led to the higher-order category of switching career path to pursue another career passion. The two respondents were FGCSs and employed (either full-time or part-time), leading to the final theme related to employed FGCSs.

In this study, we identified themes at the interpretive level to reveal the underlying ideas and assumptions within the data (Braun & Clarke, 2006). As the researchers of this study, we have actively and critically

Table 6
Comparing student agreement to the career anchors: by FGCS status.

	All (n = 70)	FGCS (n = 42)	non-FGCS (n = 22)	t	df	One-Sided p
1-Technical competence	4.2	4.26	4.18			
	(0.861)	(0.912)	(0.142)	0.401	55.316	0.345
2-Managerial competence	3.71	3.79	3.55			
	(1.038)	(1.071)	(0.205)	0.912	46.975	0.183
3-Autonomy/independence	3.96	4.12	3.82			
	(0.939)	(0.993)	(0.169)	1.317	51.661	0.097*
4-Security/stability	4.27	4.43	4.14			
	(0.962)	(0.914)	(0.211)	1.151	39.877	0.128
5-Entrepreneurship	3.5	3.71	3.23			
	(1.164)	(1.274)	(0.207)	1.704	53.563	0.047**
6-Creativity	3.46	3.67	3.18			
	(1.086)	(1.203)	(0.182)	1.866	56.276	0.034**
7-Service/dedication	3.86	3.98	3.86			
	(0.967)	(1.024)	(0.178)	0.473	50.993	0.319
8-Pure challenges	4.03	4.17	4			
	(0.884)	(0.824)	(0.186)	0.739	40.642	0.232
9-Lifestyple integration	4.07	4.17	4.09			
	(0.997)	(1.102)	(0.146)	0.338	60.076	0.368

Note. The top row for each question indicates the mean for each sample or sub-sample; numbers in parentheses are standard deviations. Six students did not indicate their FGCS status. * $p < 0.1$, ** $p < 0.05$.

examined our own assumptions and biases, recognizing that themes are a product of our engagement with the data. We understand that our experiences working with FGCS populations might have influenced our analytical lenses. We practiced reflexivity during the thematic analysis through team discussions by asking ourselves, "Are we making assumptions about a participant's words?" We discussed and debated our understanding of the data set as we reviewed the coding process and the revealed themes.

5.2. Qualitative findings

The thematic analysis revealed four themes in the career development of FGCSs and their counterparts: (1) FGCSs who are employed use their college education to change career to pursue their passion; (2) FGCSs who are employed use their college education to prepare for opening their own businesses; (3) both employed FGCSs and non-FGCSs use their college education to advance in their career; and (4) the career choices of unemployed FGCSs and non-FGCSs are motivated by their interests. Among the four themes, Themes 1 and 2 were more frequently expressed by FGCSs than non-FGCSs while Themes 3 and 4 were shared by both groups. Below we present the four themes with supporting quotes selected from the narratives of the study participants.

***Theme 1*. Employed FGCSs using a college education to change careers to pursue their passion**

FGCSs are more likely to work part-time or full-time to meet their family's financial needs. Their pre-college employment was mostly in low-paying jobs in hospitality, logistics, or customer service. A college education provides them with an opportunity to change their career path and pursue their passion. For example, two FGCSs explained their plans to use their college education to switch their career paths, from a warehouse associate to a computer programmer and from a customer service rep to sports management respectively. They expressed their career development plans in the two remarks below (*the bold emphases in all the quotes were added by the authors*):

> *[I am doing]* ***Warehouse Associate*** *Tasks. I am working towards a degree in Business Information Systems, once completed* ***I aim to move into the IT Department*** *at my current job. I aim to work in Web programming and security developing features.* (ID: 76; FGCS, business major, age 30–39, employed full-time, male, Hispanic or Latinx)

> *I am a* ***sales assistant*** *and customer service rep. After my college graduation, I am* ***interested in pursuing a career in Sports management or sports marketing.*** *I have always been passionate about sports and sporting events that bring together a community of fans. I would like to be at the forefront of organizing and planning for these events while bringing attention to these events.* (ID: 110; FGCS, business major, age 22–29, employed full-time, female, Hispanic or Latinx)

***Theme 2*. Employed FGCSs using a college education to prepare for their own business**

Some FGCSs hoped their college education would enable them to land a marketing/HR position to build skills and experience, later helping them open their own business (reflecting their entrepreneur anchor). As reflected in the remarks below, both FGCSs had a goal of eventually launching their own business, but the first career phase post-college was to land a job in the industry to accumulate the relevant expertise and experience. The two aspiring entrepreneurs explained:

> *[Currently I work in] Property Management. As a recent finance graduate, my goal is to pursue* ***a career as an administrator*** *in a commercial real estate development company. Alternatively, I aspire* ***to start my own venture*** *in this field, leveraging my financial expertise and passion for real estate.* (ID: 112; FGCS, business major, age 30–39, employed full-time, female, Black or African American)

> *[I am a] Server at a restaurant serving food and drink to patrons. I will pursue a career as a pharmaceutical representative. I will build a solid foundation of* ***experience in sales and marketing*** *so that eventually I can* ***own a business*** *teaching healing and Kundalini Yoga.* (ID: 115; FGCS, business major, age 40–49, employed full-time, female, mixed race: white and African American)

***Theme 3*. Employed students using a college education to advance in their career**

For some employed students, a college education enables them to develop competence and skills and serves as a jumping board for them to advance in their current professional fields, either in the same company or through a different company. For example, a FCGS holding an analyst position in the IT field hoped to advance to a management position in the same field. Another FGCS in an administrative position in the government sector was interested in a managerial position within the same government agency.

Table 7
Illustration of qualitative data coding and analysis process through examples.

Respondent's narrative	Relevant background	First-order codes	Higher-order categories	Final themes
Response 1: "My daily work is *TSA Screening*, essentially searching for potential terroristic threats. After graduating from college, I am interested in becoming a *financial analyst/ investment banker*. Finance has always been something that has *fascinated* me due to the many components that make this exchangeable good we call currency very valuable." (ID: 99)	FGCS; Employed full-time, Business major; Junior	(1) switching careers (from TSA screening to Finance) (2) career interest in finance	switching career path to pursue another career passion	College education enables employed FGCSs to change careers to pursue their career passion
Response 2: "In my daily work I run food to guest at a popular *burger restaurant*. After college, I want to be a *social media marketing* manager. I am already chronically online so why not make a career out of it. I also *like creating* and staying on top of trends and I feel like I can use that knowledge to help others with their business ventures." (ID: 100)	FGCS; Employed part-time, Business major; Senior	(1) switching careers (from hospitality/ restaurant to social media marketing) (2) career interest in social media marketing	To switch career path to pursue another career passion	College education enables employed FGCSs to change careers to pursue their career passion

> *I am interested in pursuing a career in IT within xx Health as a manager or some type of management position within xx. I currently work for xx as a program analyst so hopefully* ***with this degree*** *I* ***can eventually become a manager*** *in this dept.* (ID: 68; FGCS, business major, age 30–39, employed full-time, female, Black or African American)

> *[Current I am working on] admin duties such as time keeping, scheduling, supervising several employees… I am* ***interested in management level career*** *within the City of xx, such as a management analyst, and eventually become a director of a division.* (ID: 62; FGCS, public administration major, age 30–39, employed full-time, male, Hispanic or Latinx)

Similarly, non-FGCSs expressed the same intents for career advancement. Below are the remarks of two non-FGCSs:

> *I am an early childhood home visitor. After graduation, I am looking forward to* ***climbing the latter*** *[sic.]* ***at my job*** *and working in human resource department or become a supervisor or manger.* (ID: 108; non-FGCS, business major, age 30–39, employed-full time, female, Hispanic or Latinx)

> *I currently work at a government agency. The responsibilities of my position include general office/clerical tasks. Working on my degree has given me insight on how to gain success within the agency and the bachelor's degree in Public Administration will* ***help me achieve management level.*** (ID: 72; non-FGCS, public administration major, age 50–59, employed full-time, female, Black or African American)

Theme 4. ***Having no prior employment, students found their career choices motivated by their interests***

A relatively small portion of survey respondents (18.5 % of the 70 respondents) did not have any prior work experience; they were full-time college students only. Regardless of their generational status (FGCS vs. non-FGCS), these students' career orientation was motivated mainly by their interests, as reflected in the two remarks below:

> *I am pursuing a Marketing degree with the purpose to gain experience and assist smaller businesses compete with other competitors through the use of marketing. I am to begin with an entry level job as a marketing manager to eventually work towards becoming an advertising director. I am choosing this path* ***because of the love and passion*** *I have for creating ads.* (ID: 117; FGCS, business major, age 22–29, unemployed (student only), male, Hispanic or Latinx)

> *After college, I want to have my own business in the Hospitality field. I want to plan events, but specifically weddings.* ***It has always been a dream of mine*** *as I love weddings and planning and executing things.* (ID: 90; non-FGCS, business major, age 18–21, unemployed (student only), female, Black or African American)

In addition to the four themes reported above, we also analyzed the association between the entrepreneurial/creative career anchor and career trajectories but found that only a small number of respondents were interested in launching their own business, even though they rated entrepreneurship and creativity highly in their career profile. This lack of association between career anchor and career development is consistent with the result of the quantitative analysis (see Section 4.2). The few respondents who did show an entrepreneurial career goal were FGCSs majoring in business administration. This lack of more students with entrepreneurial career goals is probably because managing one's own business requires skills and knowledge about various functions of business (marketing, human resource management, etc.).

The qualitative analysis did not show that students linked their career development to the presence of AI. This is not surprising considering that at the time of the survey (May-July 2023), only 24 % of the student participants had used ChatGPT and their ChatGPT experience was very limited. As AI technologies are quickly evolving and GenAI tools such as ChatGPT are rapidly adopted by university students, we conducted a follow-up study in spring 2025 to further investigate how FGCSs and their counterparts perceive their career development in the age of AI, which allowed us to add new insights for RQ3.

6. Follow-up study and findings

6.1. Data and sample

We designed another student survey (IRB approved) and collected the data from the same research site in April 2025. Using a convenience sampling strategy, we invited nine instructors in the college to share the survey with their students in the 16 classes they were teaching. A total of 187 students (an overall response rate of 36 %) completed the 2025 survey anonymously.

Similar to the 2023 survey, most student participants were FGCSs and employed either full-time or part-time. Student participants represented diverse ethnic backgrounds with the majority being Hispanics (66.8 %), followed by Black (18.7 %), Asian or Pacific Islander (6.4 %), and White Caucasian (5.9 %). Female students comprised 47.1 % of the sample, and most students were between 18 and 39 years old. A supermajority of the participants was majoring in Business Administration (76.5 %). In general, the sample statistics are consistent with the university student demographics in terms of ethnicity and gender, while having a higher representation of FGCSs and unemployed students. Notably, in 2023, 53 of the 70 student participants (75.7 %) indicated that they had no prior experience with ChatGPT, while in 2025, only 10 of the 187 participants (5.3 %) reported that they never used ChatGPT: 44 (23.5 %) used it several times a week and 9 (4.8 %) used it daily.

The 2025 survey consisted of 30 multiple-choice or open-ended questions. The question regarding student's concern about AI's career impact was similar to that in the 2023 survey: "How do you feel when you think about your future career that might be impacted by AI?" Students were asked to rate their agreement on a 7-point scale (1 =Strongly Disagree; 7 =Strongly Agree) to the following statements: (1) I feel excited; (2) I feel happy; (3) I feel surprised; (4) I feel unease; (5) I feel confused; and (6) I feel fear. These six items were adapted from the positive and negative emotions toward AI proposed by Park and Woo (2022). The first three items represent positive emotions, and the last three items indicate negative emotions. Table 8 presents the descriptive statistics for the six emotions. We also performed *t*-tests comparing the means of the six emotions between FGCSs and non-FGCSs but found no statistical significance for any of them.

On average, the positive emotions are rated higher than the negative emotions, 4.28 vs. 3.99 of 7 points. Interestingly, the attitude "I feel surprised" (4.35) ranked the highest among the six emotions, followed by "I feel excited" and "I feel unease" equally (4.31). The rest of the emotions are "I feel happy" (4.20), "I feel fear" (3.91), and "I feel confused" (3.75).

6.2. Qualitative findings to supplement RQ3

In the 2025 follow-up study, we analyzed the completed responses by 58 participants to three open-ended questions related to their current work setting and their concerns about AI and their future career: "Please describe a required task in your current work setting or a hypothetical scenario in your desired career." "For what part of this job task do you plan to use or rely on AI? Please explain." "Do you have any additional comments and concerns about AI and your future career?" We used thematic analysis and followed the same coding process described in Section 5.1.

Table 8
Descriptive statistics of positive and negative emotions toward AI (n = 187).

	N	Minimum	Maximum	Mean	Std. Deviation
Positive Emotions:	186	1	7	4.28	1.42
1-I feel excited	187	1	7	4.31	1.73
2-I feel happy	186	1	7	4.20	1.69
3-I feel surprised	186	1	7	4.35	1.70
Negative Emotions:	186	1	7	3.99	1.40
4-I feel unease	186	1	7	4.31	1.52
5-I feel confused	186	1	7	3.75	1.57
6-I feel fear	186	1	7	3.91	1.84

The thematic analysis of 2025 data revealed four themes on the AI-associated career concerns of FGCSs and their counterparts, expanding our findings for RQ3. The four themes correspond to four types of attitudes toward AI's impact on their career: negative, neutral, positive, and a mixed attitude. Specifically, among the 58 respondents, almost 40 % expressed a negative attitude toward AI's impact on their career, followed by a neutral attitude (22.4 %), a positive attitude (19 %), and mixed feelings (19 %) (see the Appendix Table A1).

Theme 1. Negative attitude toward AI's impact on their career

Twenty-three of the 58 respondents expressed negative attitude toward AI's impact on their careers. The main reason for the negative attitude was the fear of AI replacing jobs traditionally performed by humans. This potential lack of future job opportunities caused concerns among the FGCSs and their counterparts who were motivated to pursue a college education to change or advance their careers. In addition, some respondents expressed the fear of losing human values in the age of AI. Both reasons for the negative attitude are reflected below:

> *My current job is processing payroll for all employees at the company I work in. It is* ***intimidating*** *looking towards the future with the constant worry if job opportunities will still be available for me in the workplace.* (P079; FGCS)

> I hope AI doesn't take over jobs and replace humans because as social creatures***, it will take away that value that can be detrimental to our mental well-being***. AI cannot replicate true genuine emotions of a human or for connections/relationships like a human can. (P069, non-FGCS)

Theme 2. Neutral attitude toward AI's impact on their career

Thirteen respondents did not express either a positive or a negative attitude toward AI's impact on their career. Rather, they focused on the human-oriented nature of their job. For example, one FGCS conducted new training for Marines and did not believe the career in law enforcement would likely be replaced by AI. Similarly, one non-FGCS did not believe that his job of fitting hockey skates could be replaced by AI. Their neutral attitude is reflected below:

> *I believe having a future* ***career in law enforcement would be less likely be affected by AI*** *because of how much it relies on human presence and elements. A required task in my current work setting would be to train new Marines on an upcoming field exercise so they have knowledge of what to expect.* (P082; FGCS; Employed part-time)

> *My job is mostly human based that cannot be reproduced by AI currently.* ***I am not that worried.*** *My job is Fitting Hockey Skates on people who are looking for a perfect fit for their foot's and Skating level's needs. Practically the entire process relies on human. I need to use my judgement with my eyes to determine what size a person's foot is and what kind of skates would be suitable for their skating regimen.* (P166; non-FGCS; Employed part-time)

Theme 3. Positive attitude toward AI's impact on their career

Among the 58 respondents, only eleven respondents expressed a positive view regarding AI's impact on their future careers. However, their positive viewpoints were associated with slightly different reasons. While some perceived new job opportunities from AI adoption, others considered a potential benefit of AI use as enhancing their creativity, thus advancing their future careers. Both reasons are reflected below:

I have always believed change is not something we should be scared off. With change comes new opportunities, new careers, if we don't let go of the past we cannot continue into the future. With AI some jobs may be automated but ***new jobs will be created*** *that will make us more efficient and better paying jobs. (P037; FGCS)*

AI will help me obtain new creative ideas to be able ***to grow in my future career*** *and provide me with the tools needed to succeed.* (P039; non-FGCS)

Theme 4. Mixed attitude toward AI's impact on their career

Eleven respondents expressed mixed feelings regarding AI's impact on their career. They both loved it and hated it because they believed that AI has both positive and negative aspects for their career development. Thus, they considered AI as a "double-edged sword." The mixed attitude is reflected below:

Although ***AI is a double-edged sword and has many cons and pros****, I believe the way you take advantage of it depends on how probable it is to lose your job in place of AI.* (P074; FGCS; Employed *part*-time)

My feelings towards AI are complicated *as I understand and like how AI can automate repetitive tasks, like data analytics, but at the same time, I also respect the effort it takes to reach the result yourself. I've taken some courses centered around AI … I'll likely use it (or be pressured into using AI) for my career.* (P173; non-FGCS; Employed part-time)

6.3. Attitudes toward AI and the intersection of generational status and employment

Moreover, respondents' attitudes toward AI's impact on careers appeared to be connected to the intersection of generational status and employment. More unemployed FGCSs view AI's impact on their career negatively (55.6 %), compared to unemployed non-FGCSs (40 %) (see Appendix Table A1). For example, an unemployed FGCS was afraid that the rapid growth of AI would replace entry level positions in data entry and bookkeeping and worried about becoming jobless after graduation.

I am a bit ***fearful*** *of how rapidly AI has grown and the pace at which it is developing. Accounting requires a human element and precise attention to detail so I believe it would be difficult for full replacement of accountants by AI. However, data entry positions and bookkeeping will be affecting very soon by these advancements in AI. This means less lower level jobs will be available and* ***I don't know what that means for the availability of entry level jobs as I prepare to enter the field in June 2025.*** (P005; FGCS; Unemployed)

Interestingly, more full-time employed non-FGCSs view AI's impact on their career negatively (57.1 %), compared to full-time employed FGCSs (33.3 %). In addition to the fear of job replacement, some respondents expressed the fear of losing human autonomy in the age of AI. This negative view is expressed below:

I feel that ***one day AI will take over the world****, and everyone will resort to being mind-numbing zombies, unable to create basic concepts independently. It's already started with the increased use of electronic devices in schools.* (P103; non-FGCS; Employed full-time)

In summary, the qualitative data analysis in the follow-up study revealed four types of attitudes shared by FGCS and non-FGCS, of which, positive and negative attitudes were partially consistent with those identified in a prior study (Stein et al., 2024). Moreover, we observed some differences in attitudes toward AI by generational cohorts and employment status.

7. Discussion

The main objective of this study was to understand the perceptions of the impact of AI on the careers of a vulnerable workforce, FGCSs. Taken together, the mixed-methods study provides evidence that FCGSs are more concerned about AI's impact than non-FGCSs. Moreover, students' perceptions of their career development are largely related to their employment status: in the 2023 survey, employed students commonly reported that they considered using a college education as a pathway to change to a different career, to advance in their current career, or to open their own business, while unemployed students' career orientations were motivated mainly by their interests. Compared to their non-FGCS counterparts, more FGCSs mentioned using a college education to change to different careers to pursue their passion. College education also enables them to accumulate skills and experience in a business function before launching an entrepreneurial career. The follow-up study using the 2025 survey data extended the results by pinpointing students' perceptions of the impact of AI on their career and revealed four types of attitudes toward AI (negative, positive, neutral, and mixed) among FGCSs and their counterparts.

However, this study indicates that there is a disconnect between the career planning of FGCS employees and their perceptions of the risk of AI replacement. The qualitative data analysis shows that FGCSs seek to advance their social mobility by moving from low-paying jobs to high-paying jobs via career advancement or transition. However, respondents to our survey also rated computer programmer and IT analyst positions as positions subject to a high risk of being replaced by AI. The seeming paradox is related to their AI knowledge and usage. A recent study by Bankins et al. (2024) shows that AI's effect on careers is unevenly distributed across different career stages, e.g., school, university-to-work, and established workers. Nevertheless, it is critical for both university students transitioning to work and established workers to build AI literacy and to understand the skills needed for AI use.

7.1. Theoretical contributions

Theoretically, this study contributes to career development research by examining the effect of FGCS status and career anchor on individual risk perceptions of AI's impact on careers in the context of the marginalized FGCS workforce. The quantitative analysis of the 2023 survey data demonstrated that students' FGCS status positively affects their concern about AI's impact on their career, but career anchors did not show statistically significant impacts on the risk perceptions of university students. Furthermore, the qualitative analysis revealed that FGCSs and non-FGCSs appeared to have different emphases in their career development (i.e., more FGCSs mentioned changing to a new career or launching an entrepreneurial career) although no connection was found between career anchor and individual career development.

In addition, the 2025 follow-up study contributes to our knowledge

Table 9
Individual career development and attitude toward AI's impact: by FGCS and employment status.

FGCS Status	Employment Status	Career Development	Attitude Toward AI's Impact on Career
FGCSs	Employed	Looking to advance or change careers	Positive view
	Unemployed	Looking for first job motivated by interest/passion	Negative view
Non-FGCSs	Employed	Looking to advance or change careers	Negative view
	Unemployed	Looking for first job motivated by interest/passion	Positive view

about the career development of FGCSs in the presence of AI by revealing four types of attitudes and considering these attitudes by their intersection with generational and employment status. As shown in the Appendix Table A1, a higher percentage of employed FGCSs (i.e., those with established employment) have a positive view of AI's impact, compared to unemployed FGCSs (i.e., those without employment history) and employed non-FGCSs. Table 9 compares individual career development and attitude toward AI's impact among four groups: employed FGCSs, unemployed FGCSs, employed non-FGCSs, and unemployed non-FGCSs.

Participants in this study varied in FGCS status (FGCS vs. non-FGCS) and employment status (employed vs. unemployed). They have shown different career developments after completing their college education and expressed different attitudes towards AI's impact on their careers. This is consistent with the prediction of careers theory, which posits that individuals enter different stages of their careers and may cycle back through certain stages when they encounter disruptions or changes (Lent & Brown, 2020). It is essential for individuals to prepare for career disruptions, such as in the current labor markets influenced by the rapid advancement and adoption of AI in workplaces. In this study, FGCSs who are currently employed are more likely to show a positive attitude towards AI and tend to use their college education to advance in their job ladder or to switch to a different career that reflects their passion. Meanwhile, FGCSs who do not have employment experience are more likely to have a negative attitude towards AI's impacts but still insist on pursuing a career to fulfill their career passions. This finding is consistent with previous studies showing that each career stage has distinct individual needs at a given time, necessitating tailored workforce development services and guidance (Bankins et al., 2024).

Our study of career development by FGCSs and their counterparts suggests that career paths are no longer linear but a series of transitions and learning cycles and both FGCS status and employment status have influences on individuals' career paths. The findings extend the careers theory (e.g., Lent & Brown, 2020) by not only accounting career stages (e.g., employment status) but also employees' FGCS status and motivation to pursue college education.

7.2. Practical implications

The study provides practical implications for the career development of FGCSs in the age of AI. Based on the study results, we suggest three sets of career development interventions to help FGCSs cope with the perceived risks of job replacement by AI.

First, we suggest interventions focusing on motivating FGCSs to pursue career advancement or transitions to achieve upward social mobility. For example, our study revealed that students considered analyst positions at a high risk of being replaced by AI, and FGCSs perceived higher levels of such risks compared to their non-FGCS counterparts. Thus, FGCSs who seek to advance to a managerial position may pivot their competence development to human-centered competencies such as communication and social interaction skills, which are often required in management positions. Interventions such as managerial communication bootcamps could help students practice public speaking and active listening, learn how to handle conflicts and difficult conversations, build cross-functional collaborations, manage team emotions, etc.

Second, we suggest revisiting and updating curricula by incorporating the latest AI technologies and providing FGCSs with the technical skills and abilities they need in order to adapt to future work integrated with AI. At a minimum, it is essential for FGCSs to become proficient users of GenAI tools (such as ChatGPT) to enhance their experiential learning (Sun & Deng, 2025) and work productivity while mitigating their anxiety and fear about the negative impact of AI on the future of their work and careers. As revealed in our study, students with prior ChatGPT experience were less concerned about AI's impact on their career. As AI is becoming an integral part of organizational work, we believe strong technical skills and familiarity with GenAI tools will become an important success factor in FGCSs' career advancements in the age of AI. Thus, we recommend intervention strategies such as AI literacy and upskilling workshops in partnership with tech companies to equip FGCSs with in-demand tech skills to remain competitive in the evolving job market.

Finally, our study highlights the need for educational institutions, faculty, and career service staff to become more aware of the career-related challenges faced by FGCSs in the rapidly changing workplace transformed by AI. Our study calls for reimagining programs and policies that better prepare FGCSs to start or advance their professional careers. Prior research has offered insights into supporting FGCSs during the early stages of their careers by focusing on their individual self-efficacy, outcome expectations, and personal goals (Deng et al., 2022; Olson, 2014; Tate et al., 2015). However, in the age of AI, focusing on factors at the individual level is no longer sufficient. As revealed in the quantitative analysis of our study, individual career anchors are not associated with FGCSs' concerns about AI's impact on their career. Thus, FGCSs should be encouraged to understand the extent to which an occupation in a profession is likely to be replaced by AI and to actively acquire knowledge and develop the AI-related skills required for that occupation. As suggested by the differences in attitudes about AI's impact (Table 9), when universities offer FGCSs and their counterparts career support services and workforce development guidance, those services and guidelines need to be tailored to the unique needs of FGCSs and their counterparts with different employment status (e.g., employed vs. unemployed).

8. Conclusion

Using mixed methods, this exploratory study examined the influence of students' FGCS status and career anchor on their concerns about AI's impact on their career as well as revealed students' perceptions of AI in their career development. Through analyzing quantitative data collected from 70 students in a business college at a minority-serving public university in the United States in 2023, the study showed that a student's FGCS status is positively associated with their concern about AI's impact on their career, while prior ChatGPT experience is negatively associated with this concern. Regarding the 19 business and public administration positions, FGCSs perceived a higher risk of job replacement by AI than did their non-FGCS counterparts. However, we did not find evidence that a student's career anchor affects their concerns about AI's career impact. The qualitative analysis further revealed four themes in the career development of FGCSs and their counterparts.

Using another survey conducted in 2025, our follow-up study enhanced the qualitative analysis findings by focusing on participants' concerns about AI's impact on their career. Our findings suggest that university students' perceptions of AI's impact and their career development not only differed by their generational status but also depended on their employment status (employed vs. unemployed).

The study has its limitations, and the findings should be interpreted with caution. One limitation is that the small convenience sample collected from one type of higher education institution may affect the generalizability of the findings. For future research, large, more diverse and random samples are suggested to advance the quantitative analysis to improve the generalizability of the findings. Another limitation is that the data used in this study did not consider one's risk-taking behavior, which may affect students' perceptions about the threat of AI to their careers or potential careers. Risk-taking behavior may vary greatly among the diverse body of FGCSs (e.g., by gender, ethnicity, or field of study), which was not accounted for in this study.

One promising research area is to deepen our understanding of AI literacy and the development of occupation-specific, AI-relevant skills to enhance competencies among FGCS students and established workers. Without a well-defined understanding of these skill requirements, workers may find it challenging to plan and execute necessary actions to remain competitive in the labor market (Bankins et al., 2024). Another promising direction is to examine risk-taking behavior, as it is considered one of the behavioral skills important for career success. Risk taking refers to the undertaking of actions that have the potential for positive outcomes but also for negative outcomes. Not only is risk taking seen as career enhancing, but it is also a stereotypically gendered trait, one that continues to be seen as a core aspect of the masculine stereotype and as being incompatible with the feminine stereotype (Ellemers, 2018). In this regard, future research on risk taking differences by FGCS status and gender could bring new insights. Finally, studies exploring the role of mentorship in helping FGCSs navigate AI-related challenges are likely to yield further insights. As prior research has shown, mentorship has played a significant, positive role in FGCSs' educational journey: it has been a successful intervention for helping FGCSs navigate the post-secondary process (Glass, 2023); both faculty mentoring and peer mentoring have increased FGCSs' confidence, belongingness, and involvement in studying a biomedical engineering (Ahmed et al., 2021).

The rapid development and implementation of AI have important implications for risk and shaping the future of work and society more broadly. Important questions still need to be answered in the space of AI and labor markets. Scholars across disciplines have expressed the concern that in the long run, AI can further marginalize the already marginalized populations such that they end up in an even less equitable position (Arora et al., 2023) and have emphasized the importance of considering the consequences of AI in advance, rather than historically (Bailey & Barley, 2020). To follow the calls of the scholars, our study makes an initial attempt to explore the potential negative consequences of AI on the career development of a marginalized population, FGCSs, and provide insights into understanding and overcoming AI-related career challenges experienced by them. We hope that this mixed-methods study on AI and the career trajectories of this marginalized population will serve to attract and excite more scholars to conduct this line of research, one which matters to the future of work and labor markets in the AI era.

CRediT authorship contribution statement

Xuefei (Nancy) Deng: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rui Sun:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This study was partially supported by a 2024-25 Research, Scholarship, and Creative Activity Award through California State University, Dominguez Hills.

Appendix

Table A1
Distribution of attitude toward AI by generational cohorts and employment status

Attitude toward AI	FGCS				Non-FGCS (CGCS)				Total
	Employed FT	Employed PT	Not employed	Total	Employed FT	Employed PT	Not employed	Total	
By Frequency Count									
Negative	3	6	5	14	4	3	2	9	23
Neutral	2	3	1	6	2	4	1	7	13
Positive	3	5		8	1		2	3	11
Mixed feeling	1	2	3	6		5		5	11
Total	9	16	9	34	7	12	5	24	58
By Percentage of Column Total									
Negative	33.3 %	37.5 %	55.6 %	41.2 %	57.1 %	25.0 %	40.0 %	37.5 %	39.7 %
Neutral	22.2 %	18.8 %	11.1 %	17.6 %	28.6 %	33.3 %	20.0 %	29.2 %	22.4 %
Positive	33.3 %	31.3 %	0.0 %	23.5 %	14.3 %	0.0 %	40.0 %	12.5 %	19.0 %
Mixed feeling	11.1 %	12.5 %	33.3 %	17.6 %	0.0 %	41.7 %	0.0 %	20.8 %	19.0 %
Total	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %

Data Availability

The data that has been used is confidential.

References

ACT Inc. (2014). Understanding the underserved learners: The condition of STEM 2014. https://www.act.org/content/dam/act/unsecured/documents/STEM-Underserved-Learner.pdf.

Ahmed, M., Muldoon, T. J., & Elsaadany, M. (2021). Employing faculty, peer mentoring, and coaching to increase the self-confidence and belongingness of first-generation college students in biomedical engineering. *Journal of Biomechanical Engineering, 143*(12), Article 121001.

Arora, A., Barrett, M., Lee, E., Oborn, E., & Prince, K. (2023). Risk and the future of AI: Algorithmic bias, data colonialism, and marginalization. *Information and Organization, 33*(3), Article 100478.

Bailey, D. E., & Barley, S. R. (2020). Beyond design and use: How scholars should study intelligent technologies. *Information and Organization, 30*(2), Article 100286.

Bankins, S., Jooss, S., Restubog, S. L. D., Marrone, M., Ocampo, A. C., & Shoss, M. (2024). Navigating career stages in the age of artificial intelligence: A systematic interdisciplinary review and agenda for future research. *Journal of Vocational Behavior*, Article 104011.

Belanche, D., Casaló, L. V., Flavián, C., & Schepers, J. (2020). Service robot implementation: A theoretical framework and research agenda. *The Service Industries Journal, 40*(3-4), 203–225.

Belanche, D., Belk, R. W., Casaló, L. V., & Flavián, C. (2024). The dark side of artificial intelligence in services. *The Service Industries Journal, In Press*.

Belk, R. W., Belanche, D., & Flavián, C. (2023). Key concepts in artificial intelligence and technologies 4.0 in services. In *Service Business, 17* pp. 1–9).

Bowen, J. A., & Watson, C. E. (2024). *Teaching with AI. Science, 381* pp. 187–192).

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. In *Qualitative Research in Psychology, 3* pp. 77–101).

Chang, C. L.-H., Chen, V., Klein, G., & Jiang, J. J. (2011). Information system personnel career anchor changes leading to career changes. *European Journal of Information Systems, 20*(1), 103–117.

Chen, X., & Carroll, C. D. (2005). *First-generation students in postsecondary education: A look at their college transcripts (NCES 2005-171)*. Washington, DC: U.S. Department of Education, National Center for Education Statistics. *United States Government Printing Office*.

Clarke, V., Braun, V., & Hayfiel, N. (2015). Thematic analysis. In: Smith J (ed) *Qualitative psychology: a practical guide to research methods,* 3rd ed. *Sage, London*, 222–248.

Dika, S. L., & D'Amico, M. M. (2016). Early experiences and integration in the persistence of first-generation college students in STEM and non-STEM majors. *Journal of Research in Science Teaching, 53*(3), 368–383.

Deng, X., Zaza, S., & Armstrong, D. J. (2022). What motivates first-generation college students to consider an IT career? An integrative perspective, 10.17705/1CAIS.05203 *Communications of the Association for Information Systems, 50*. https://doi.org/10.17705/1CAIS.05203.

Dwivedi, Y. K., Hughes, L., Ismagilova, E., Aarts, G., Coombs, C., Crick, T., & Williams, M. D. (2021). Artificial Intelligence (AI): Multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research, practice and policy. *International Journal of Information Management, 57*, Article 101994.

Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., & Wright, R. (2023). "So what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management, 71*, Article 102642.

Ellemers, N. (2018). Gender stereotypes. *Annu. Rev Psychol, 69*, 275–298.

Engle, J., & Tinto, V. (2008). Beyond access: College success for low-income, first-generation students. *Washington, DC: Pell Institute for the Study of Opportunity in Higher Education*.

Feldman, D. C., & Bolino, M. C. (1996). Careers Within Careers: Reconceptualizing the Nature of Career Anchors and Their Consequences. *Human Resource Management Review, 6*, 89–112.

Feuerriegel, S., Hartmann, J., Janiesch, C., & Zschech, P. (2024). Generative AI. *Business and Information Systems Engineering, 66*, 111–126. https://doi.org/10.1007/s12599-023-00834-7

Flavián, C., Belk, R. W., Belanche, D., & Casaló, L. V. (2024). Automated social presence in AI: Avoiding consumer psychological tensions to improve service value. *Journal of Business Research, 175*, Article 114545.

Fu, R., Huang, Y., & Singh, P. (2021). Crowds, lending, machine, and bias. *Information Systems Research, 32*(1), 72–92.

Glass, L. E. (2023). Social capital and first-generation college students: Examining the relationship between mentoring and college enrollment. *Education and Urban Society, 55*(2), 143–174.

Hirudayaraj, M., & McLean, G. N. (2018). First-generation college graduates: A phenomenological exploration of their transition experiences into the corporate sector. *European Journal of Training and Development, 42*(1/2), 91–109.

Kinder, M., de Souza Briggs, X., Muro, M., & Liu, S. (2024). Generative AI, the American worker, and the future of work. *Brookings Report, 10*, 2024. ⟨https://www.brookings.edu/articles/generative-ai-the-american-worker-and-the-future-of-work/⟩ (October).

Kshetri, N., Dwivedi, Y. K., Davenport, T. H., & Panteli, N. (2023). Generative artificial intelligence in marketing: Applications, opportunities, challenges, and research agenda. *International Journal of Information Management*, Article 102716.

Lent, R. W., & Brown, S. D. (2020). Career decision making, fast and slow: Toward an integrative model of intervention for sustainable career choice. *Journal of Vocational Behavior, 120*(103), 448. https://doi.org/10.1016/j.jvb.2020.103448

Lohfink, M. M., & Paulsen, M. B. (2005). Comparing the determinants of persistence for first-generation and continuing-generation students. *Journal of College Student Development, 46*(4), 409–428.

Mingers, J. (2001). Combining is research methods: Towards a pluralist methodology. *Information Systems Research, 12*, 240–259.

Mingers, J., Mutch, A., & Willcocks, L. (2013). Critical realism in information systems research. *MIS Quarterly, 37*, 795–802.

Nassauer, S. (2025). Walmart CEO Issues Wake-Up Call: 'AI Is Going to Change Literally Every Job', September 26, 2025. *Wall Street Journal*. ⟨https://www.wsj.com/tech/ai/walmart-ceo-doug-mcmillon-ai-job-losses-dbaca3aa⟩.

Nunez, A. M., & Cuccaro-Alamin, S. (1998). First-generation students: Undergraduates whose parents never enrolled in postsecondary education (NCES No. 98-082). Statistical Analysis Report, Postsecondary Education Descriptive Analysis Reports. *Washington, DC: United States Department of Education*.

Olson, J. S. (2014). Opportunities, obstacles, and options: First-generation college graduates and social cognitive career theory. *Journal of Career Development, 41*(3), 199–217.

Palmer, A. (2025a). *Amazon CEO Jassy says AI Willich lead to 'fewer People Doing some of the jobs' that Getting automated, CNBC*. ⟨https://www.cnbc.com/2025/06/30/amazon-ceo-says-ai-will-mean-fewer-people-do-jobs-that-get-automated.html⟩. , June 30.

Palmer, A. (2025b). *Amazon Laying off about 14,000 Corporate workers as it invests more in Ai CNBC*. ⟨https://www.cnbc.com/2025/10/28/amazon-layoffs-corporate-workers-ai.html⟩. , October 28.

Pandya, S. S., & Wang, J. (2024). Artificial intelligence in career development: a scoping review. *Human Resource Development International. 27*(3), 324–344.

Park, J., & Woo, S. E. (2022). Who likes artificial intelligence? Personality predictors of attitudes toward artificial intelligence. *The Journal of Psychology, 156*(1), 68–94.

Redford, J., & Hoyer, K. (2017). First generation and continuing-generation college students: A comparison of high school and postsecondary experiences (NCES 2018-009), U.S. Department of Education. *Washington, DC: National Center for Education Statistics*.

Repko, M. (2025). Target cuts 1,800 corporate jobs in its first major layoffs in a decade. October 23, 2025. *CNBC*. ⟨https://www.cnbc.com/2025/10/23/target-layoffs-corporate-jobs-sales-slump.html⟩.

Schein, E. H. (1978). *Career dynamics: Matching individual and organizational needs, 6834*. Addison-Wesley.

Schein, E. H. (1985). *Career anchors: Discovering your real values*. San Diego: University Associates.

Schein, E. H. (1996). Career anchors revisited: Implications for career development in the 21st century. *The Academy of Management Executive, 10*(4), 80–88.

Schein, E. H. (2006). *Career anchors: Discovering Your Real Values*. San Francisco: Pfeiffer.

Schepers, J., Belanche, D., Casaló, L. V., & Flavián, C. (2022). How smart should a service robot be? *Journal of Service Research, 25*(4), 565–582.

Stein, J. P., Messingschlager, T., Gnambs, T., Hutmacher, F., & Appel, M. (2024). Attitudes towards AI: measurement and associations with personality. *Scientific Reports, 14*(1), 2909.

Subin, S. (2025). AI is doing up to 50% of the work at Salesforce. *CEO Marcius Benioff says*. ⟨https://www.cnbc.com/2025/06/26/ai-salesforce-benioff.html⟩. June 26, 2025, *CNBC*.

Sun, R., & Deng, X. (2025). Using ChatGPT to enhance experiential learning: An exploratory study of generative ai in higher education. *Journal of Information Systems Education, 36*(1), 53–64.

Taylor, J., & Joshi, K. D. (2019). Joining the crowd: The career anchors of information technology workers participating in crowdsourcing. *Information Systems Journal, 29*(3), 641–673.

Tate, K. A., Caperton, W., Kaiser, D., Pruitt, N. T., White, H., & Hall, E. (2015). An exploration of first-generation college students' career development beliefs and experiences. *Journal of Career Development, 42*(4), 294–310.

The Lancet Regional Health – Europe (2023). Editorial: Embracing generative AI in health care. The Lancet Regional Health – Europe, 30, 100677.

Toyokawa, T., & DeWald, C. (2020). Perceived career barriers and career decidedness of first-generation college students. *The Career Development Quarterly, 68*(4), 332–347.

US Department of Education (2021). List of postsecondary institutions enrolling populations with significant percentages of undergraduate minority students. ⟨https://www.doi.gov/pmb/eeo/doi-minority-serving-institutions-program⟩. Accessed 14 Oct 2024.

Van Doorn, J., Mende, M., Noble, S. M., Hulland, J., Ostrom, A. L., Grewal, D., & Petersen, J. A. (2017). Domo arigato Mr. Roboto: Emergence of automated social presence in organizational frontlines and customers' service experiences. *Journal of Service Research, 20*(1), 43–58. https://doi.org/10.1177/1094670516679272

Wils, T., Wils, L., & Tremblay, M. (2014). Revisiting the career anchor model: A proposition and an empirical investigation of a new model of career value structure. *Relations Industries, 69*(4), 813–838.

Wirtz, J., Hofmeister, J., Chew, P. Y. P., & Ding, X. D. (2023). Digital service technologies, service robots, AI, and the strategic pathways to cost-effective service excellence. *Service Industries Journal, 43*(15-16), 1173–1196.

Wong, I. A., Lian, Q. L., & Sun, D. (2023). Autonomous travel decision-making: An early glimpse into ChatGPT and generative AI. *Journal of Hospitality and Tourism Management, 56*, 253–263.